



COURSE SYLLABUS						
Cours e Code	INTE 351	Course Title	Systems Analysis and Design		Course Credit	3 UNITS
Course Description		The course is a study of the different phases of information systems development. Discussions will concentrate on the initiation, analysis and design of a system. This course strikes a balance between the theoretical and applied aspects of systems analysis, presenting state-of-the-art systems procedures, methodology and software. Skills that will be acquired are: understanding the current system, determining the problems and their root causes and providing solutions to these problems. These skills are applied by allowing the students to experience analyzing and designing a 'live' system for an outside client before they do it in a real-world project.				
Pre-Requisites		COMP 20093	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



INSTITUTIONAL LEARNING OUTCOMES (ILO)	As a polytechnic state university, PUP shall develop its students to possess: 1. Critical and Creative Thinking. Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world. 2. Effective Communication. Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances. 3. Strong Service Orientation. Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors. 4. Adept and Responsible Use or Development of Technology. Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills. 5. Passion for Lifelong Learning. Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.
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	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11. Produce technologies for commercialization of livelihood improvement.



PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
	Apply knowledge of computing, science, and mathematics appropriate to the discipline.	✓								
	Analyze complex problems, and identify and define the computing requirements appropriate to its solution.	✓								
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.	✓								
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.	✓								



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	Integrate IT-based solutions into the user environment effectively.	✓										
	Assist in the creation of an effective IT project plan.	✓										
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession. (ILO 4)				✓							
	Understand best practices and standards and their applications.			✓								
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.							✓				
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.		✓									
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.											✓
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.								✓			
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.					✓						



COURSE LEARNING OUTCOMES (CLO)	At the end of this course, the students are expected to:	Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Describe Object Oriented Analysis and Design concepts and apply them to solve problems.								
	Prepare Object Oriented Analysis and Design documents for a given problem using Unified Modeling Language.								



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1	Identify and describe the Object Oriented concepts	Introduction to Object Oriented Concepts: Classes, Objects, Abstraction, Inheritance, Polymorphism, Encapsulation, Message Sending, Associations, Aggregation and Composition	Completed Review Questions; Functional Specifications Documentation; Midterm Examination; Group Defenses and Consultations; Oral Defense		Quizzes, Midterm Examination, Group Defenses, Oral Defense					
2	Define object modelling and explain its benefits Define UML and its various types of diagrams	Object Oriented Analysis and Modeling: Introduction to Object Oriented Analysis; Introduction to Unified Modeling Language (UML)								



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3	Recognize the benefits of a software development process Identify the different Process Models available Describe the stages of RUP	Software Development Process: Introduction; Process Models; Rational Unified Process (RUP)							
4-5	Describe the benefits of Use-Case Modeling Define Actors, use cases and use-case relationships Identify and describe the steps for preparing a use-case model	Creating Use Case Diagrams: Actors, Use Cases, Use Case Relationships; Process of Requirements Use-Case Modeling; Constructing the Analysis Use-Case Model							
6-7	Recognize concepts and notations for finding Objects and Classes Identify Stereotypes and Packages Draw Package/Class/Object/Composite Structure Diagrams Illustrate Association and Aggregation relationships	Identifying Classes, Packages and drawing Class/Object/Composite Structure Diagrams: Visualizing a Class; Associations; Inheritance and Generalization; Package Diagrams; Aggregations/Composites; Interfaces and Realizations							
8	Differentiate object class types	Object Oriented Design and Modeling Using UML: Types of							

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	Identify design relationships Define visibility and its three levels Describe object-oriented design activities and reusability/design patterns	Object Classes; Design Relationships; Attribute/Method Visibility; Object Responsibilities; Object Reusability and Design Patterns.							
9	MIDTERM EXAMINATION								
10	Draw state transition diagrams for classes with significant dynamic behaviour	Working with State Diagrams: Introduction; UML 2.0 State Diagrams; Importance of State Diagrams							
11-12	Add scenarios describing how use cases are realized as interactions among objects Describe scenarios	Discovering Object Interactions: Sequence Diagrams; Communication Diagrams							



	via sequence and communication diagrams								
13	Model use case activities using Activity Diagrams Identify and draw swimlanes	Working with Activity Diagrams: Modeling use case activities; swimlanes							
14	Describe component/deployment diagrams and the physical architecture they show Describe Composite Structure, Interaction Overview, Timing,	Component and Deployment Diagrams; New Diagrams in UML 2.x: Composite Structure, Interaction Overview, Timing, Profile Diagrams							



	and Profile Diagrams								
15-17	Draw the UML diagrams for a given project to be developed	Project Development							
18		Project Presentation, Evaluation and feedback							



LIST OF REFERENCES

BOOKS:

Shelly (2009), Systems Analysis and Design

Satzinger (2009), Systems Analysis and Design in a Changing World, 5th ed.

Booch, G. (2007), Object Oriented Analysis and Design with Applications, 3rd ed.

Cashman, S. (2005), Systems Analysis and Design, 6th ed.

Booch, Jacobson, Rumbaugh (2005), The Unified Modeling Language User Guide, 2nd ed., Addison-Wesley

Hoffer, J., et al. (2004), Modern System Analysis and Design, 3rd edition.

Kendall and Kendall (2004), Systems Analysis and Design, 6th ed.

Whitten, Bentley, Dittman (2004), Systems Analysis Design Methods

Kendall & Kendall (2002), Systems Analysis and Design, 5th ed., Pearson Education Asia

Cashman & Thomas (2001), Systems Analysis and Design, 4th ed.

Valacich, George, Hoffer (2001), Essentials of Systems Analysis and Design

Harris (1999), Systems Analysis and Design for Small Enterprise

Whitten & Bentley (1999), Systems Analysis and Design Methods, 4th ed., Tata MacGraw-Hill

Hoffer et al. (1999), Modern Systems Analysis and Design, 2nd ed., Addison-Wesley

Hawryszkiewicz (1998), Introduction to System Analysis and Design, 4th ed., Prentice-Hall

IBM, Essentials of Visual Modeling, IBM Philippines, Inc.

IBM, Mastering Requirements Management Using Use-Case, IBM Philippines, Inc.

WEBSITE:

<http://www.uml.org>

Extended Readings may be assigned by the professor. Instructional Media: videos related to the topics listed above.



CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Refer to General Rules (right column) for attendance, honesty, and submission policies that applied to this course.	• Assignments and research projects/report works will be given throughout the semester. Such requirements shall be due as announced in class. Late submission shall be penalized with grade deductions (5% per day) or shall no longer be accepted, depending on the subject facilitator's discretion. Assignments and exercises are designed to assist you in understanding the materials presented in class, and to prepare you for the exams. • Students are required to attend classes regularly, including possible make-up classes. The student will be held liable for all topics covered and assignments made during his/her absence. The university guidelines on attendance and tardiness will be implemented. Every meeting, an attendance sheet will be passed around — be sure to sign it. • Academic honesty should be practiced at all times. Any evidence of copying or cheating on any course work may result in a failing grade for all parties involved. • Students are advised to keep graded work until the semester has ended. • Contents of the syllabus are subject to modification with notification. • Cell phone, radio or other listening devices are not allowed to be used inside lecture and laboratory rooms to prevent any distractive interruption of the class activity. • No food, drinks, cigarettes, nor children are allowed inside the lecture and laboratory rooms. • Only officially enrolled students are allowed inside the lecture



	or laboratory room. • Withdrawal and dropping of the subject should be done in a. accordance with existing university policies and guidelines. s
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	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.	
COURSE REQUIREMENT/S	GRADING SYSTEM	
- Quizzes (at least 3) - Class Recitation - Group Consultation - Assignments & Exercises - Group Defenses (at least 3) - Oral Defense	Mid Term	Final Term
	Class Standing	Class Standing
	Quizzes (at least 3)	group defence (at least 3): 50% oral defence: 50:
	class recitation	
	group consultation	
assignments & exercises		
attendance		
Midterm Examinations		
Final Grade = (Midterm Grade + Final Term Grade) / 2		
CLASS INFORMATION		FACULTY INFORMATION
Course, Year and Section:		Name of Faculty:
Day and Time:		Consultation Time:
Room:		Office Tel. No./ Mobile Phone No.
School Year:	Semester:	E-Mail Address:



General Rules:

- Aside from what is prescribed in the student handbook, the following are the professor’s additional house rules:
1. Assignments and research projects/report works will be given throughout the semester. Such requirements shall be due as announced in class. Late submission shall be penalized with grade deductions (5% per day) or shall no longer be accepted, depending on the subject facilitator’s discretion. Assignments and exercises are designed to assist you in understanding the materials presented in class, and to prepare you for the exams.
 2. Students are required to attend classes regularly, including possible make-up classes. The student will be held liable for all topics covered and assignments made during his/her absence. The university guidelines on attendance and tardiness will be implemented. Every meeting, an attendance sheet will be passed around. Be sure to sign it.
 3. Academic honesty should be practiced at all times. Any evidence of copying or cheating on any course work may result in a failing grade for all parties involved.
 4. Students are advised to keep graded work until the semester has ended.
 5. Contents of the syllabus are subject to modification with notification.
 6. Cell phone, radio or other listening devices are not allowed to be used inside lecture and laboratory rooms to prevent any distractive interruption of the class activity.
 7. No food, drinks, cigarettes, nor children are allowed inside the lecture and laboratory rooms.
 8. Only officially enrolled students are allowed inside the lecture or laboratory room.
 9. Withdrawal and dropping of the subject should be done in accordance with existing university policies and guidelines.

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
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